

Centre Number	Candidate Number	Candidate Name
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NAMIBIA SENIOR SECONDARY CERTIFICATE

BIOLOGY ADVANCED SUBSIDIARY LEVEL

8223/2

PAPER 2

1 hour 15 minutes

Marks 60

2023

Additional Material: Non-programmable calculator

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the Question Paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- You may use a non-programmable calculator.
- Do not use correction fluid.

- Answer **all** questions.
- The number of marks is given in brackets [] at the end of each question or part question.
- You will lose marks if you do not show your working or if you do not use appropriate units.

For Examiner's use

1	
2	
3	
4	
5	
Total	

Marker	
Checker	

This document consists of **15** printed pages and **1** blank page.



Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 In a study done to find a suitable location for a biodiversity hiking trail in the Huibes Conservancy, a total of 120 plant species in two different habitats were surveyed. The habitats were randomly selected. Habitat one was along a riverbed and habitat two was in rocky terrain.

In habitat 1, two straight line transects of 2 km each were laid down, 10 m apart. Every plant found at each 10 m interval along the transect was identified and recorded.

In habitat 2, two line transects of 2 km each were laid down 10 m apart and plants occurring at each 10 m interval were identified and recorded.

The results for habitat 1 are shown in Table 1.1, and the results for habitat 2 in Table 1.2.

Table 1.1

species	number	$\frac{n}{N}$	$\left(\frac{n}{N}\right)^2$
<i>Dichrostachys cineria</i>	16		
<i>Boscia foetida</i>	2		
<i>Acacia melifera</i>	8		
<i>Catophractes alexandri</i>	2		
<i>Boscia albitrunca</i>	23		
<i>Acacia senegal</i>	1		
<i>Kleina longiflora</i>	21		
<i>Parkinsonia africana</i>	4		
<i>Lycium villosum</i>	7		
<i>Calicorema capitata</i>	8		
<i>Rhigozun trichotomum</i>	8		
Total (N)			
D			

Table 1.2

species	number	$\frac{n}{N}$	$\left(\frac{n}{N}\right)^2$
<i>Ziziphus mucronata</i>	1		
<i>Albizia anthelmintica</i>	1		
<i>Lycium villosum</i>	3		
<i>Dichrostachys cinerea</i>	2		
<i>Boscia foetida</i>	1		
<i>Acacia mellifera</i>	1		
<i>Acacia senegal</i>	6		
<i>Catophractes alexandri</i>	2		
<i>Boscia albitrunca</i>	1		
<i>Combretum apiculatum</i>	1		
Total (N)			
D			

- (a) Use the information in each of the tables to calculate Simpson's Index of

Diversity (D) for each habitat using the formula $D = 1 - \sum \left(\frac{n}{N}\right)^2$ where

n = number of individuals of each species present in the samples

N = the total number of all individuals of all species

Show all your working and write your answers to four decimal places in Table 1.1 and Table 1.2, respectively.

[4]

- (b) The Simpson's Index of Diversity is an indication of the biodiversity in an area.

Explain the meaning of the term *biodiversity*.

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[2]

(c) Compare the biodiversity of the two habitats.

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[2]

(d) Discuss the need to maintain biodiversity.

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[2]

[Total: 10]

- 2 The genetic material of some viruses is found in their RNA molecules, with which they can directly produce their proteins in the cytoplasm of host cells.

(a) Explain how the structure of RNA differs from the structure of DNA.

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[3]

- (b) DNA carries the genetic information of cells and is found in the nucleus. During the S-phase of the cell cycle, DNA is replicated semi-conservatively before new cells are produced.

State **two** features of DNA and explain how these are important in the process of semi-conservative replication of DNA.

1

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2

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[2]

(c) Fig. 2.1 shows a part of the processes taking place during protein synthesis.

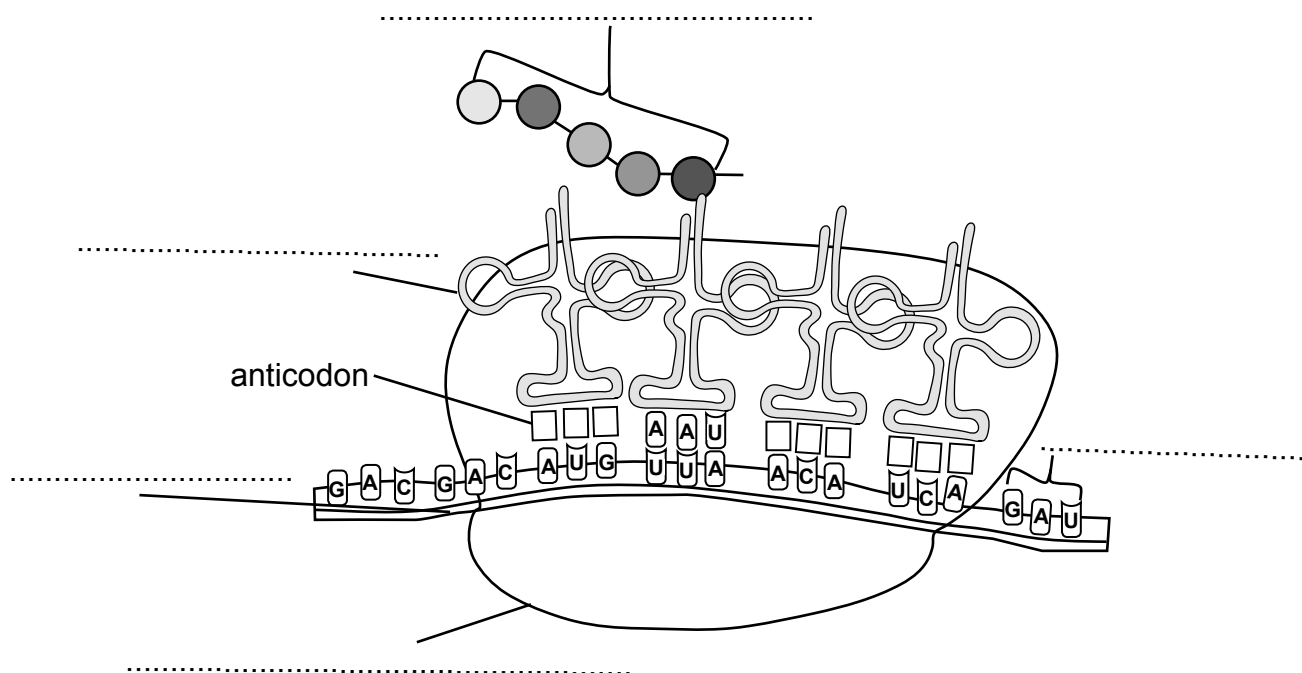


Fig. 2.1

- (i) Complete Fig. 2.1 by labelling the five parts on the answer lines. [3]
- (ii) Fill in all the missing anticodons on the tRNA molecules using the codons provided in Fig. 2.1. [3]
- (iii) Table 2.1 shows the base triplets that code for the four amino acids shown in Fig. 2.1.

Table 2.1

RNA code	AUG	UUA	ACA	GAU
amino acid	Methionine	Leucine	Threonine	Aspartic acid

Describe how any two amino acids differ from each other.

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[1]

[Total: 12]

- 3 Several different types of vaccines have been developed for viral diseases. These include vaccines that contain inactivated or weakened viruses or protein-based vaccines which make use of harmless fragments of proteins that mimic the actual virus.

Fig. 3.1 shows a cell from the immune system engulfing viruses.

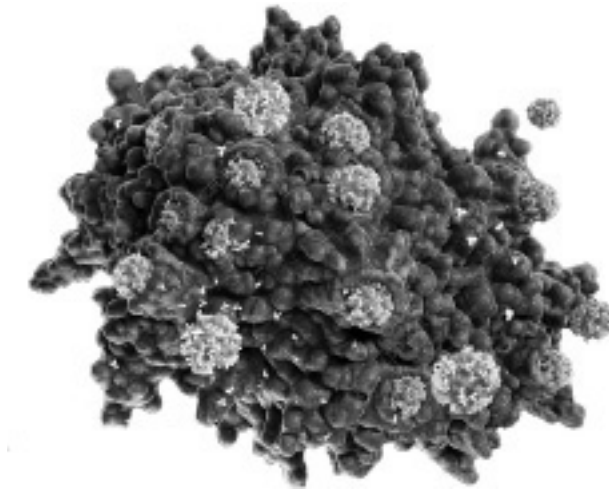


Fig. 3.1

- (a) (i) Name the type of cell shown in Fig. 3.1.

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[1]

- (ii) Fig. 3.2 shows part of a cell membrane of the cell shown in Fig. 3.1.

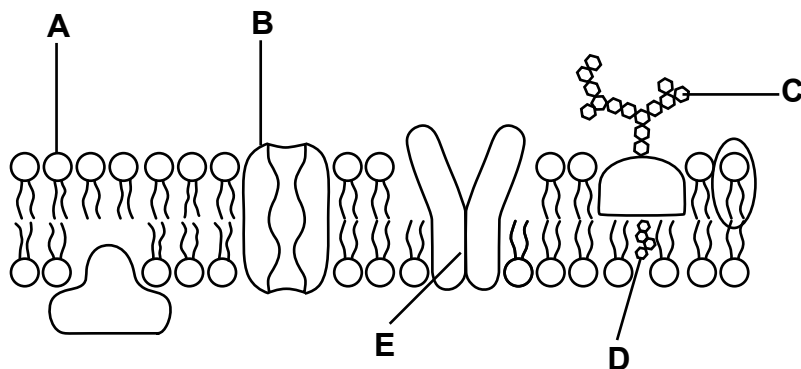


Fig. 3.2

State which part of the cell membrane enables this cell to recognize the virus which it is engulfing.

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[1]

- (iii) Describe all the events taking place after the cell has engulfed the virus until the production of antibodies in response to the vaccine.

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[5]

- (iv) Vaccination against a disease stimulates the formation of memory cells. Explain the role of memory cells in the defence of the body against a pathogen like a virus.

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[3]

- (b) Tuberculosis (TB) in humans is caused by the pathogen *Mycobacterium tuberculosis*. In most cases this disease involves the lungs, as is the case with some viruses. TB is treated by making use of different antibiotics, such as isoniazid and rifampin. Isoniazid interferes with cell wall synthesis in *M. tuberculosis* and rifampin inhibits bacterial RNA polymerase.

- (i) Explain why the antibiotic isoniazid leads to the death of the bacterial cells.

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[2]

- (ii) Explain why the antibiotic isoniazid cannot be used to treat viruses.

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[1]

- (c) Table 3.1 shows the trends in consumption of antibiotics in some European countries, expressed as DDD (defined daily doses) per 1000 inhabitants per day.

Table 3.1

Country	year					percentage change between 2012 and 2016
	2012	2013	2014	2015	2016	
Netherlands	11.3	10.8	10.6	10.7	10.4	
Finland	19.5	18.3	18.1	17.2	16.5	-15.39
Greece	32.5	32.2	35.1	36.1	36.3	-11.69

- (i) Calculate the percentage change for the Netherlands between 2012 and 2016 and give your answer to 2 d.p. in Table 3.1.

Show your working.

[1]

- (ii) In the light of increasing antibiotic resistance, overuse of antibiotics is a cause for concern. State the countries listed in Table 3.1 which show an awareness of this situation in their use of antibiotics.

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[1]

- (iii) Explain why antibiotic resistance is a cause for concern.

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[1]

- (iv) Microbiologists can test bacteria for antibiotic resistance. This can be done by soaking paper discs in antibiotics and placing them on Petri dishes containing agar with bacterial cultures.

An example of such an experiment is shown in Fig. 3.3.

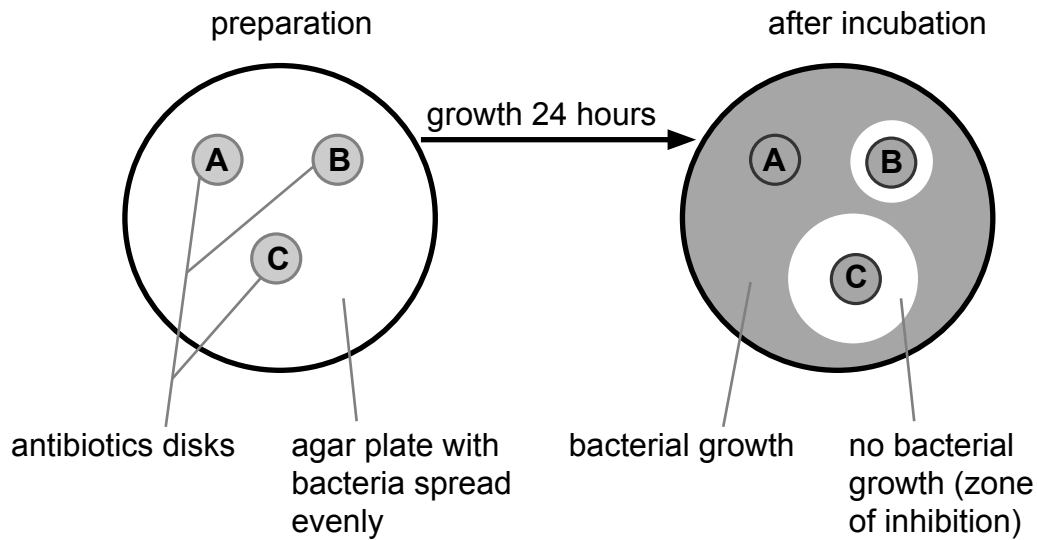


Fig. 3.3

State and explain a conclusion which could be reached for each antibiotic by looking at the Petri dish after 24 hours.

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[3]

[Total:19]

- 4 Fig. 4.1 shows lung tissue taken from the same person, but in different regions. **A** shows normal tissue and **B** shows tissue affected by COPD.

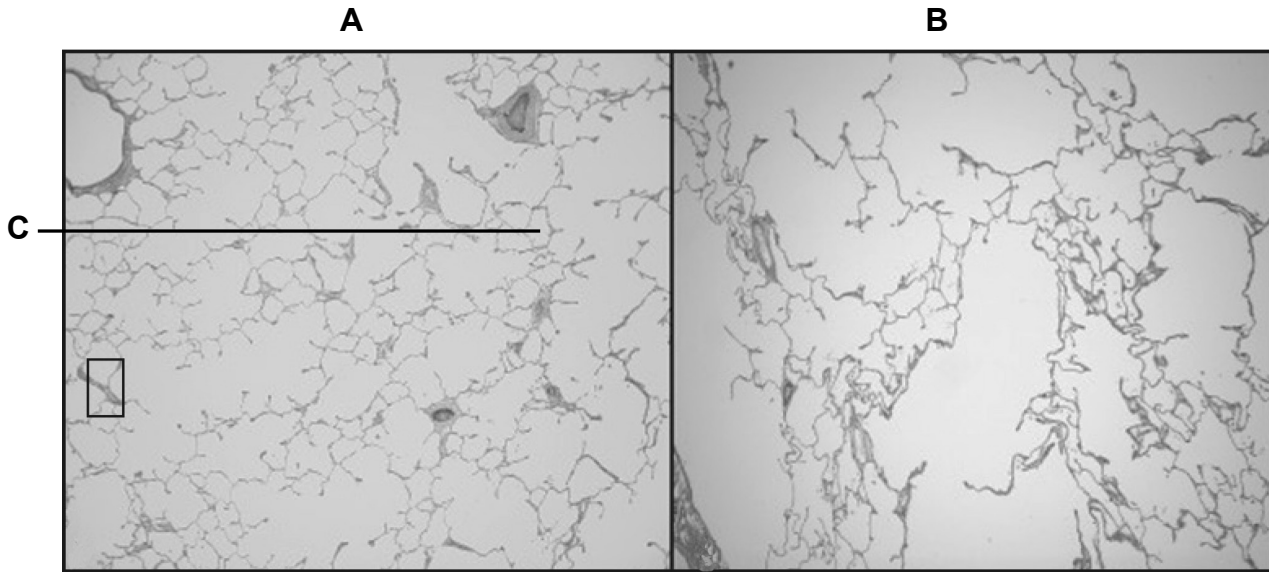


Fig. 4.1

- (a) (i) Name the structures in the lung which are labelled **C** in Fig. 4.1 **A** and state their function in the human body.

name

function.....

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[2]

- (ii) Discuss how the squamous epithelial cells, which line the structures labelled **C** in Fig. 4.1 **A**, affect the rate of diffusion.

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[2]

(b) Fig. 4.2 shows an enlarged section of the tissue as shown in the rectangular block in Fig. 4.1 **A**.

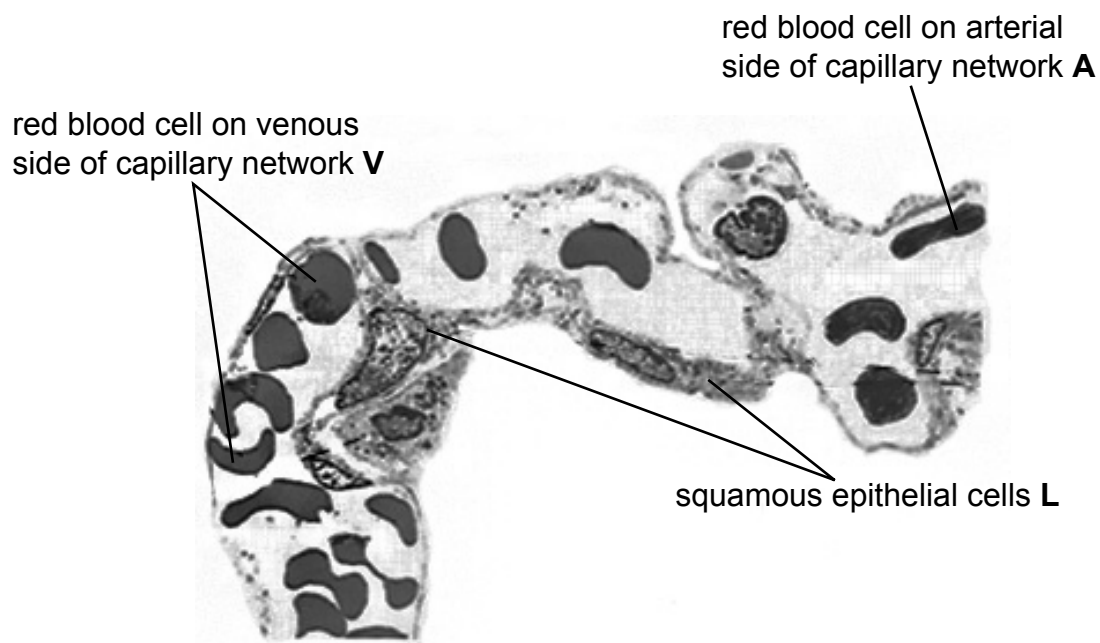


Fig. 4.2

Table 4.1 shows the partial pressures of oxygen (pO_2) and carbon dioxide (pCO_2) at **A**, **L** and **V**, as shown in Fig. 4.2.

Table 4.1

partial pressure	locations in lung tissue as shown in Fig. 4.2		
	L	V	A
pO_2 / kPa	13.87	13.87	5.33
pCO_2 / kPa	5.33	5.55	6.00

Using data from Table 4.1 and referring to Fig. 4.2, describe the exchange of gases as blood flows from **A** to **V**.

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[4]

(c) Explain the effect of COPD, as seen in Fig. 4.1 B, on gas exchange.

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[3]

[Total: 11]

5 Enzymes can be immobilized for use in industry.

(a) Explain what is meant by immobilizing enzymes.

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[2]

(b) Suggest **two** reasons why it is more efficient to immobilize the enzymes rather than adding the enzymes directly to the solution in which the reaction should take place.

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2

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[2]

(c) Fig. 5.1 shows the apparatus which can be used to investigate the use of immobilized enzymes. A solution, containing the substrate for the enzymes, was poured into a separating funnel which contained an immobilized enzyme. The investigation was done using two different enzyme concentration.

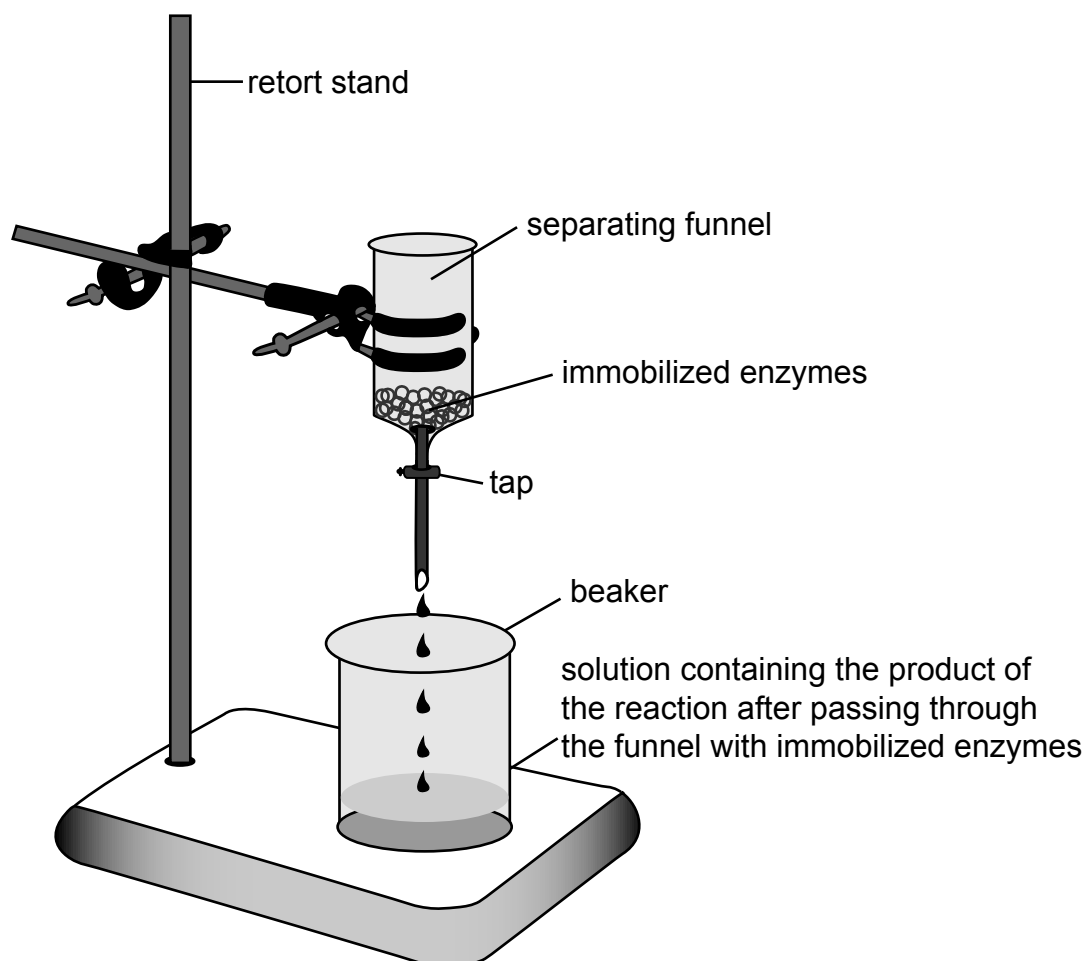


Fig. 5.1

The solution which passed through the separating funnel was collected after a fixed time period and the concentrations of the substrate as well as the product were measured. Five learners carried out the investigation and each learner did the investigation twice.

Table 5.1 shows the results obtained.

Table 5.1

	enzyme concentration				enzyme concentration			
	0.2 g dm ⁻³		0.4 g dm ⁻³		0.2 g dm ⁻³		0.4 g dm ⁻³	
	substrate concentration / gdm ⁻³				product concentration / gdm ⁻³			
	repeat 1	repeat 2	repeat 1	repeat 2	repeat 1	repeat 2	repeat 1	repeat 2
Learner 1	24	26	14	13	32	33	60	64
Learner 2	25	22	12	12	34	39	60	63
Learner 3	22	23	10	13	35	32	59	61
Learner 4	18	24	11	12	34	33	62	68
Learner 5	25	28	13	18	30	32	65	64

The learners came to the following conclusion from this investigation:

Doubling the enzyme concentration doubled the rate of the enzyme-catalysed reaction.

- (i) Use data in the table to state **one** way in which the data supports the conclusion.

.....
 [1]

- (ii) State **two** ways in which the reliability of the results could be improved.

1

 2
 [2]

- (d) State **one** example from industry where immobilized enzymes are used.

..... [1]

[Total: 8]

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